Bayes' Rule

$$P(A \wedge B) = P(A)\ P(B|A)$$

$$P(A \wedge B) = P(B)\ P(A|B)$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Or, $P(A \cap B) = P(A|B)\ P(B)$ — (i)

$$P(B|A) = \frac{P(B \cap A)}{P(A)}$$

Or, $P(A \cap B) = P(B|A)\ P(A)$ — (ii)

Comparing (i) & (ii) $\Rightarrow$

$$P(B|A) \cdot P(A) = P(A|B) \cdot P(B)$$

Or, $$P(B|A) = \frac{P(A|B)\ P(B)}{P(A)}$$

$$P(A|B) = \frac{P(B|A)\ P(A)}{P(B)}$$

Bayes' Theorem

$$P(A|B) = \frac{P(B|A)\,P(A)}{P(B)}$$

- $P(A|B)$ → Posterior probability
- $P(B|A)$ → Likelihood
- $P(A)$ → Prior probability
- $P(B)$ → Evidence

$$P(W|X) = \frac{P(X|W)\,P(W)}{P(X)}$$

X - feature
W - Class

$$P(\text{Disease} \mid \text{Symptom}) = \frac{P(\text{Symptom} \mid \text{Effect}) \cdot P(\text{Disease})}{P(\text{Symptom})}$$

A doctor knows that the disease meningitis causes the patient to have a stiff neck, say, 50% of the time and whiplash 80% of the time. The doctor also knows some unconditional facts: the prior probability of a patient having meningitis is 1/50000, and having whiplash is 1/10000. If a patient comes to a doctor which treatment will be given to him and why?

→ Given,

$P(S|M) = 50\% = 0.5$

↳ Posterior

$P(S|W) = 80\% = 0.8$

↳ Prior

$P(M) = \frac{1}{50000}$

$P(W) = \frac{1}{10000}$

$$P(M|S) = \frac{P(S|M)\,P(M)}{P(S)} \quad \text{—— (i)}$$

$$P(W|S) = \frac{P(S|W)\,P(W)}{P(S)} \quad \text{—— (ii)}$$

(i) ÷ (ii) ⇒

$$\frac{P(M|S)}{P(W|S)} = \frac{\frac{P(S|M)\cdot P(M)}{P(S)}}{\frac{P(S|W)\cdot P(W)}{P(S)}}$$

$$= \frac{P(S|M)\,P(M)}{P(S|W)\,P(W)}$$

$$= \frac{0.5 \cdot \frac{1}{50000}}{0.8 \cdot \frac{1}{10000}}$$

$$= 0.125$$

$\Rightarrow P(MIS) = 0.125 \times P(WIS)$ | Here, $P(WIS) > P(MIS)$

That is, whiplash is more than meningitis in terms of probability. So, the doctor will treatment for whiplash.

A doctor knows that pneumonia causes a fever 95% of the time. She knows that if a person is selected randomly from the population, there is a 10^{-7} chance of the person having pneumonia. 1 in 100 people suffer from fever. You go to the doctor complaining about the symptom of having a fever (evidence). What is the probability that pneumonia is the cause of this symptom (hypothesis)?

→ Given,

$$P(F|Pn) = 95\% = 0.95$$

$$P(Pn) = 10^{-7}$$

$$P(F) = \frac{1}{100}$$

$$P(Pn|F) = ?$$

$$P(Pn|F) = \frac{P(F|Pn)\,P(Pn)}{P(F)}$$

$$= \frac{0.95 \times 10^{-7}}{0.01}$$

$$= 9.5 \times 10^{-6}$$

Bayes Theorem

$$P(A|B) = \frac{P(B|A)\,P(A)}{P(B)}$$

likelihood ↑ , Priore →

$P(A|B)$ ↓ Posteriore

$P(B)$ ↓ evidence

$$P(\omega|x) = \frac{P(x|\omega)\,P(\omega)}{P(x)}$$

x - feature

ω - class

Example 01

$$P(SN|M) = 0.5$$

$$P(M) = 1/50000$$

$$P(SN) = 1/20$$

$$P(M|SN) = \frac{P(SN|M)\,P(M)}{P(SN)}$$

$$= \frac{0.5 \times 1/50000}{1/20} = 0.0002$$

$$P(\neg M|SN) = 1 - 0.0002$$

$$= 0.9998$$

Naive Bayes Classifier ← Supervised machine learning algorithm

$P(H|X)$ → posterior probability

X ← evidence

H ← hypothesis

NBC : Training Dataset

Target column / Output ↓

age	income	student	credit_rating	buys_computer
<=30	high	no	fair	no
<=30	high	no	excellent	no
31...40	high	no	fair	yes
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
31...40	low	yes	excellent	yes
<=30	medium	no	fair	no
<=30	low	yes	fair	yes
>40	medium	yes	fair	yes
<=30	medium	yes	excellent	yes
31...40	medium	no	excellent	yes
31...40	high	yes	fair	yes
>40	medium	no	excellent	no

Class:

C1: buys_computer = 'yes'

C2: buys_computer = 'no'

$$P(C_1|X) = \frac{P(X|C_1)\,P(C_1)}{P(X)}$$

So,

$$P(C_i|X) = \frac{P(X|C_i)\,P(C_i)}{P(X)}$$

Data Sample:

X = (age <= 30, income = medium, student = yes, credit_rating = fair, buys_computer = ?)

→ P(Ci):

$$P(\text{buys_computer} = \text{"yes"}) = \frac{9}{14} = 0.643$$

$$P(\text{buys_computer} = \text{"no"}) = \frac{5}{14} = 0.357$$

Compute P(X | Ci) for each class:

$$P(\text{age} = \text{"<= 30"} \mid \text{buys_computer} = \text{"yes"}) = \frac{2}{9} = 0.222$$

$$P(\text{age} = \text{"<= 30"} \mid \text{buys_computer} = \text{"no"}) = \frac{3}{5} = 0.6$$

$$P(\text{income} = \text{"medium"} \mid \text{buys_computer} = \text{"yes"}) = \frac{4}{9} = 0.444$$

$$P(\text{income} = \text{"medium"} \mid \text{buys_computer} = \text{"no"}) = \frac{2}{5} = 0.4$$

$$P(\text{student} = \text{"yes"} \mid \text{buys_computer} = \text{"yes"}) = \frac{6}{9} = 0.667$$

$$P(\text{student} = \text{"yes"} \mid \text{buys_computer} = \text{"no"}) = \frac{1}{5} = 0.2$$

//

P(credit_rating = "fair" | buys_computer = "yes") = $\frac{6}{9}$ = 0.667

P(credit_rating = "fair" | buys_computer = "no") = $\frac{2}{5}$ = 0.4

- P(X | C_i) :

  P(X | buys_computer = "yes")

  = 0.222 * 0.444 * 0.667 * 0.667

  = 0.044

  P(X | buys_computer = "no")

  = 0.6 * 0.4 * 0.2 * 0.4

  = 0.019

- P(X | C_i) * P(C_i) :

  P(X | buys_computer = "yes") * P(buys_computer = "yes")

  = 0.044 * 0.643

  = 0.028

  P(X | buys_computer = "no") * P(buys_computer = "no")

  = 0.019 * 0.357

  = 0.0067

Therefore, X belongs to class ("buys_computer = yes")

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Example 03

$P(\text{Play} = \text{Yes}) = \frac{9}{14}$

$P(\text{Play} = \text{No}) = \frac{5}{14}$

$P(\text{Outlook} = \text{Sunny} \mid \text{Yes}) = \frac{2}{9}$

$P(\text{Outlook} = \text{Sunny} \mid \text{No}) = \frac{3}{5}$

$P(\text{Temp} = \text{Cool} \mid \text{Yes}) = \frac{3}{9}$

$P(\text{Temp} = \text{Cool} \mid \text{No}) = \frac{1}{5}$

$P(\text{Hum} = \text{High} \mid \text{Yes}) = \frac{3}{9}$

$P(\text{Hum} = \text{High} \mid \text{No}) = \frac{4}{5}$

$P(W = \text{Strong} \mid \text{Yes}) = \frac{3}{9}$

$P(W = \text{Strong} \mid \text{No}) = \frac{3}{5}$

P xNo is greater

Suppose $X = (\text{Outlook} = \text{Sunny}, \text{Temp} = \text{Cool}, \text{Humidity} = \text{High}, \text{Wind} = \text{Strong})$

$x_{YES} = P(\text{Sunny} \mid \text{Yes}) \times P(\text{Cool} \mid \text{Yes}) \times P(\text{High} \mid \text{Yes}) \times P(\text{Strong} \mid \text{Yes}) \times P(\text{Yes})$

$= 2/9 \times 3/9 \times 3/9 \times 3/9 \times 9/14 = 0.00529$

$x_{NO} = 3/5 \times 1/5 \times 4/5 \times 3/5 \times 5/14 = 0.0205$

$x_{YES} * P(\text{Play} \mid \text{Yes}) = \frac{9}{14} \times 0.00529 = 0.0034$

$N_o = \frac{5}{14} \times 0.0205 = 0.0073$

GOOD LUCK

Example 02

$P(\text{Evade} = \text{Yes}) = 3/10$

$P(\text{Evade} = \text{No}) = 7/10$

$P(\text{Married} \mid \text{Evade} = \text{Yes}) = 0/3 = 0$

$P(\text{Married} \mid \text{Evade} = \text{No}) = 4/7$

$P(\text{Refund} = \text{N} \mid \text{Evade} = \text{Y}) = 3/3 = 1$

$P(\text{Refund} = \text{N} \mid \text{Evade} = \text{N}) = 4/7$

For Income, when Evade = Y

$$\text{Mean}, M = \frac{95 + 85 + 90}{3} = 90$$

$$\text{Variance}, \sigma^2 = \frac{\Sigma (x_i - \mu)^2}{n - 1} = \frac{(95-90)^2 + (85-90)^2 + (90-90)^2}{3-1}$$

$$= 25$$

For income, when Evade = N

$$\text{Mean}, \mu = \frac{125 + 100 + 70 + 120 + 60 + 220 + 75}{7} = 110$$

$$\text{Variance}, \sigma^2 = \frac{\Sigma (x_i - \mu)^2}{n - 1}$$

$$= \frac{(125-110)^2 + (100-110)^2 + (70-110)^2 + (120-110)^2 + (60-110)^2 + (220-110)^2 + (75-110)^2}{7-1}$$

$$= 2975$$

$$P(\text{Inc} = 120k \mid \text{Evade} = Y)$$

$$= \frac{1}{2\pi\sigma^2} e^{\frac{-(x-\mu)^2}{2\sigma^2}}$$

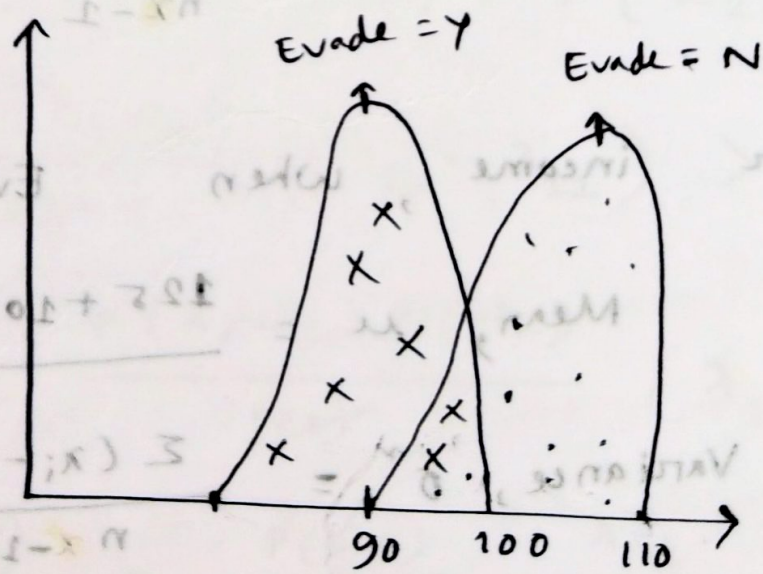
$$= \frac{1}{2\pi 25} e^{\frac{-(120-90)^2}{2 \cdot 25}} = 9.69 \times 10^{-11}$$

Feature / Class

$$P(\text{Inc} = 120k \mid \text{Evade} = N)$$

$$= \frac{1}{2\pi\sigma^2} e^{\frac{-(x-\mu)^2}{2\sigma^2}}$$

$$= \frac{1}{2\pi 2975} e^{\frac{-(120-110)^2}{2 \cdot 2975}} = 5.26 \times 10^{-5}$$



$P(\text{Married}, \text{Inc} = 120k, \text{Refund} = N \mid \text{Evade} = Y)$

$= 0 \times 9.69 \times 10^{-11} \times 1 \times \frac{3}{10} = 0$

$P(\text{Married}, \text{Inc} = 120k, \text{Refund} = N \mid \text{Evade} = N)$

$= \frac{4}{7} \times 5.26 \times 10^{-5} \times \frac{4}{7} \times \frac{7}{10}$

$= 1.20 \times 10^{-5}$

Since, $P(\text{Married}, 120k, N \mid \text{Evade} = Y) = 0$

$\therefore P(\text{Evade} = N \mid \text{Married}, 120k, N) >$

$P(\text{Evade} = Y \mid \text{Married}, 120k, N)$

So, we can trust him.

Discriminant function for Gaussian Density when the covariance matrix is variance times the identity matrix.

Decision Boundary

$$g_1(x) = g_2(x)$$

$$g_1(x) = \ln P(x/\omega_i) + \ln P(\omega_i)$$

Now,

$$N(\mu, \Sigma) = \frac{e^{-1/2(x-\mu)^t (x-\mu)\Sigma^{-1}}}{(2\pi)^{d/2} |\Sigma|^{1/2}}$$

If $P(x/\omega_i) \sim N(\mu, \Sigma_i)$ then,

$$g(x) = \ln \frac{e^{-1/2(x-\mu)^t \Sigma^{-1}(x-\mu)}}{(2\pi)^{d/2}|\Sigma|^{1/2}} + \ln P(\omega_i)$$

$$= \ln\left(e^{-1/2(x-\mu)^t \Sigma^{-1}(x-\mu)}\right) - \ln\left((2\pi)^{d/2}|\Sigma|^{1/2}\right) + \ln P(\omega_i)$$

$$= -1/2 (x-\mu_i)^t \Sigma^{-1} (x-\mu_i) - \ln\left((2\pi)^{d/2}\right) - \ln\left(|\Sigma|^{1/2}\right) + \ln P(\omega_i)$$

$$= -1/2 (x-\mu_i)^t \Sigma^{-1} (x-\mu_i) - \underbrace{\frac{d}{2}\ln 2\pi - \frac{1}{2}\ln|\Sigma| + \ln P(\omega_i)}_{\text{disregarded constant part}}$$

Case I :

$$\Sigma_i = \sigma^2 I$$

$$g_i(x) = -1/2\, \Sigma^{-1} (x-\mu_i)^t (x-\mu_i) + \ln P(\omega_1)$$

$$= -1/2 \frac{\|x-\mu_i\|^2}{\Sigma} + \ln P(\omega_1)$$

$$= -\frac{\|x-\mu_i\|^2}{2\sigma^2 I} + \ln P(\omega_1)$$

$$[w.w^T = \|w\|^2]$$

Error
Loss
Risk

TOPIC NAME : _____ DAY : _____

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α - action
λ - loss
R - Risk

For taking action α_i :

$$R(\alpha_i/x) = \sum_{j=1}^{c} \lambda(\alpha_i/\omega_j) P(\omega_j/x)$$

α_1 = decide ω_1

α_2 = " ω_2

	ω_1	ω_2
α_1	11	12
α_2	21	22

$$\lambda_{ij} = \begin{cases} 0 ; & \text{if } i = j \\ 1 ; & \text{if } i \neq j \end{cases}$$

(ω_1)	(ω_2)
$\lambda_{12} = 1$	$\lambda_{11} = 0$
$\lambda_{21} = 1$	$\lambda_{22} = 0$

$$R(\alpha_1/x) = \lambda_{11} P(\omega_1/x) + \lambda_{12} P(\omega_2/x)$$

$$R(\alpha_2/x) = \lambda_{21} P(\omega_1/x) + \lambda_{22} P(\omega_2/x)$$

Min^m Risk Decision Rule:

Decide ω_1 if $R(\alpha_1/x) < R(\alpha_2/x)$

$\Rightarrow \lambda(\alpha_1/\omega_1) P(\omega_1/x) + \lambda(\alpha_1/\omega_2) P(\omega_2/x) <$
$\lambda(\alpha_2/\omega_1) P(\omega_1/x) + \lambda(\alpha_2/\omega_2) P(\omega_2/x)$

$\Rightarrow \lambda_{11} P(\omega_1/x) + \lambda_{12} P(\omega_2/x) <$
$\lambda_{21} P(\omega_1/x) + \lambda_{22} P(\omega_2/x)$

$\Rightarrow P(\omega_1/x)(\lambda_{11} - \lambda_{21}) < P(\omega_2/x)(\lambda_{12} - \lambda_{22})$

$\Rightarrow P(\omega_1/x)(\lambda_{21} - \lambda_{11}) > P(\omega_2/x)(\lambda_{22} - \lambda_{12})$

$$\Rightarrow \frac{P(x/\omega_1)\,P(\omega_1)}{\cancel{P(x)}} (\lambda_{21} - \lambda_{11}) > \frac{P(x/\omega_2)\,P(\omega_2)}{\cancel{P(x)}} (\lambda_{12} - \lambda_{22})$$

$$\Rightarrow \frac{P(x/\omega_1)}{P(x/\omega_2)} > \frac{P(\omega_2)}{P(\omega_1)} \quad \frac{\lambda_{12} - \lambda_{22}}{\lambda_{21} - \lambda_{11}} \qquad (i = j)$$

$$\Rightarrow \frac{P(x/\omega_1)}{P(x/\omega_2)} > \frac{P(\omega_2)}{P(\omega_1)} \quad \frac{1-0}{1-0}$$

$$\therefore \frac{P(x/\omega_1)}{P(x/\omega_2)} > \frac{P(\omega_2)}{P(\omega_1)}$$

Minimum Error-rate:

$$R(\alpha_i|x) = \sum_{j=1}^{c} \lambda(\alpha_i|\omega_j) \cdot P(\omega_j|x)$$

$$= \sum_{j \neq i} \lambda(\alpha_i|\omega_j) \cdot P(\omega_j|x) + \lambda(\alpha_i|\omega_i) \cdot P(\omega_i|x)$$

$$= \sum_{j \neq i} P(\omega_j|x) \qquad \left[\because \lambda(\alpha_i|\omega_j) \to 1,\ \lambda(\alpha_i|\omega_i) \to 0\right]$$

$$= 1 - P(\omega_i|x)$$

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$$R(\alpha_1|x) = \sum_{j=1}^{2} \lambda(\alpha_1|\omega_j) \cdot P(\omega_j|x)$$

$$= \lambda(\alpha_1|\omega_1) \cdot P(\omega_1|x) + \lambda(\alpha_1|\omega_2) \cdot P(\omega_2|x)$$

$$= 5 \times 0.01 + 60 \times 0.99$$

$$= 59.45$$

$$R(\alpha_2|x) = \sum_{j=1}^{2} \lambda(\alpha_2|\omega_j) \cdot P(\omega_j|x)$$

$$= \lambda(\alpha_2|\omega_1) \cdot P(\omega_1|x) + \lambda(\alpha_2|\omega_2) \cdot P(\omega_2|x)$$

$$= 50 \times 0.01 + 3 \times 0.99$$

$$= 3.47$$

$$R(\alpha_3|x) = \sum_{j=1}^{2} \lambda(\alpha_3|\omega_j) \cdot P(\omega_j|x)$$

$$= \lambda(\alpha_3|\omega_1) \cdot P(\omega_1|x) + \lambda(\alpha_3|\omega_2) \cdot P(\omega_2|x)$$

$$= 10000 \times 0.01 + 0 \times 0.99$$

$$= 100$$

Linear Discriminant Functions

$$g(x) = 5x_1 - 2x_2 + 3$$

$$w = \begin{bmatrix} 5 \\ -2 \end{bmatrix}$$

r for a new point $(1, 1)$

→ We know,

$$r = \frac{g(x)}{\|w\|}$$

$$= \frac{5x_1 - 2x_2 + 3}{\sqrt{5^2 + (-2)^2}}$$

$$= \frac{5.1 - 2.1 + 3}{\sqrt{29}}$$

$$= \frac{6}{\sqrt{29}}$$

For the discriminant function $g(x) = 5x_1 - 2x_2 + 3$ with $w = [5 \ -2]$, explain how the distance (r) of a point $(1, 1)$ from the decision surface is determined.

Maximum Likelihood Event (MLE)

Maximum - A - Posterior Probability (MAP)

MAP

$$P(w_i|x) = \frac{P(x|w_i) \cdot P(w_i)}{P(x)}$$

ML

$$P(x|w_i) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

[x = feature]

✍ To determine the decision boundary between Salmon and Sea-bass apply MAP classifier. Given, the number of Salmon is twice that of Sea-bass, and their lengths follow Gaussian distributions with standard deviations of 1 inch for Salmon and 2 inches for Sea-bass.

→ Given,

Salmon (w_1)

where,

Mean, $\mu_1 = 5$

Standard deviation, $\sigma_1 = 1$

Variance, $\sigma_1^2 = 1^2 = 1$

Now,

Sea-bass (w_2)

where,

Mean, $\mu_2 = 10$

Standard deviation, $\sigma_2 = 2$

Variance, $\sigma_2^2 = 2^2 = 4$

Prior probabilities:

$P(w_1) = \frac{2}{3}$

$P(w_2) = \frac{1}{3}$

Total = 2+1 = 3

Salmon takes 2 parts

Seabass ,, 1 part

Assume,

300 fish

Salmon = 200

Sea-bass = 100

$P(\text{Salmon}) = \frac{200}{300} = \frac{2}{3}$

$P(\text{Sea-bass}) = \frac{100}{300} = \frac{1}{3}$

MAP boundary is where posteriors are equal:

$$P(x|\omega_1)\,P(\omega_1) = P(x|\omega_2)\,P(\omega_2)$$

1D Gaussian,

$$P(x|\omega_i) = \frac{1}{\sqrt{2\pi\sigma_i^2}} \exp\left(-\frac{(x-\mu_i)^2}{2\sigma_i^2}\right)$$

$$\Rightarrow \frac{1}{\sqrt{2\pi\sigma_i^2}} \exp\left(-\frac{(x-\mu_1)^2}{2\sigma_1^2}\right) P_1 = \frac{1}{\sqrt{2\pi\sigma_2^2}} \exp\left(-\frac{(x-\mu_2)^2}{2\sigma_2^2}\right) P_2$$

$$\Rightarrow \frac{1}{\sqrt{2\pi(1)}}\, e^{\frac{-(x-5)^2}{2}} P_1 = \frac{1}{\sqrt{2\pi(4)}}\, e^{\frac{-(x-10)^2}{2\cdot 4}} P_2$$

$$\Rightarrow \frac{1}{\sqrt{2\pi}}\, e^{\frac{-(x-5)^2}{2}} P_1 = \frac{1}{2\sqrt{2\pi}}\, e^{\frac{-(x-10)^2}{8}} P_2$$

$$\Rightarrow P_1\, e^{\frac{-(x-5)^2}{2}} = \frac{P_2}{2}\, e^{\frac{-(x-10)^2}{8}}$$ [Cancelled $\frac{1}{\sqrt{2\pi}}$ from both sides]

$$\Rightarrow 2P_1\, e^{\frac{-(x-5)^2}{2}} = P_2\, e^{\frac{-(x-10)^2}{8}}$$ [Multiplied both sides by 2]

$$\Rightarrow \frac{2P_1}{P_2}\, e^{\frac{-(x-5)^2}{2}} = e^{\frac{-(x-10)^2}{8}}$$ [Divided by P_2]

$$\Rightarrow \ln\left(\frac{2P_1}{P_2}\right) + \ln\left(e^{\frac{-(x-5)^2}{2}}\right) = \ln\left(e^{\frac{-(x-10)^2}{8}}\right)$$

[Taken 'ln' of both sides]

$$\Rightarrow \ln\left(\frac{2P_1}{P_2}\right) - \frac{(x-5)^2}{2} = -\frac{(x-10)^2}{8} \quad [\because \ln(e^a) = a]$$

$$\Rightarrow \frac{(x-10)^2}{8} - \frac{(x-5)^2}{2} + \ln\left(\frac{2P_1}{P_2}\right) = 0$$

$$\Rightarrow \frac{x^2-20x+100}{8} - \frac{x^2-10x+25}{2} + \ln\left(\frac{2P_1}{P_2}\right) = 0$$

$$\Rightarrow \frac{x^2-20x+100}{8} - \frac{4x^2-40x+100}{8} + \ln\left(\frac{2P_1}{P_2}\right) = 0$$

$$\Rightarrow \frac{x^2-20x+100-4x^2+40x-100}{8} + \ln\left(\frac{2P_1}{P_2}\right) = 0$$

$$\Rightarrow \frac{-3x^2+20x}{8} + \ln\left(\frac{2P_1}{P_2}\right) = 0$$

$$\Rightarrow -3x^2 + 20x + 8\ln\left(\frac{2P_1}{P_2}\right) = 0$$

$$\Rightarrow 3x^2 - 20x - 8\ln\left(\frac{2P_1}{P_2}\right) = 0$$

$$\Rightarrow 3x^2 - 20x - 8\ln\left(\frac{2\cdot\frac{2}{3}}{\frac{1}{3}}\right) = 0$$

$$\Rightarrow 3x^2 - 20x - 8\ln(4) = 0$$

$$\Rightarrow \quad x = \frac{20 \pm \sqrt{400 + 96\ln(4)}}{6}$$

$$\therefore \quad x = \frac{20 \pm 23.09}{6}$$

[Quadratic formula]

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$ax^2 + bx + c = 0$$

$$\Rightarrow x = \frac{43.09}{6}$$

$$\therefore x \approx 7.18$$

$$x = \frac{-3.09}{6}$$

$$\therefore x \approx -0.52$$

[fish length cannot be negative]

Decision boundary:

$$x \approx 7.18 \text{ inches}$$

MAP rule:

If $x < 7.18 \Rightarrow$ Salmon

If $x > 7.18 \Rightarrow$ Sea-bass

How increasing feature-space dimensionality is affects the observed model performance.

☒ Applying MLE (Maximum Likelihood) classifier, we ignore the priors (ignore Salmon is twice Sea-bass)

→ The decision boundary:

$$P(x|\omega_1) = P(x|\omega_2)$$

Given,

Salmon, ω_1 : $\mu_1 = 5$, $\sigma_1 = 1$

Sea-bass, ω_2 : $\mu_2 = 10$, $\sigma_2 = 2$

$$\frac{1}{\sigma_1} \exp\left(-\frac{(x-\mu_1)^2}{2\sigma_1^2}\right) = \frac{1}{\sigma_2} \exp\left(-\frac{(x-\mu_2)^2}{2\sigma_2^2}\right)$$ ⌜ exponent

$$\Rightarrow \frac{1}{1} \exp\left(-\frac{(x-5)^2}{2}\right) = \frac{1}{2} \exp\left(-\frac{(x-10)^2}{8}\right)$$

$$\Rightarrow -\frac{(x-5)^2}{2} = \ln\frac{1}{2} - \frac{(x-10)^2}{8}$$ [∵ Taken ln both sides]

$$\Rightarrow \frac{(x-10)^2}{8} - \frac{(x-5)^2}{2} + \ln(2) = 0$$

$$\Rightarrow 3x^2 - 20x - 8\ln(2) = 0$$

$$\therefore x = \frac{20 \pm \sqrt{400 + 96 \ln(2)}}{6}$$

$$= \frac{20 \pm \sqrt{466.54}}{6}$$

$$= \frac{20 \pm 21.60}{6}$$

$$\therefore x \approx 6.93 \text{ inches}$$

MLE decision rule:

$$x < 6.93 \Rightarrow \text{Salmon}$$

$$x > 6.93 \Rightarrow \text{Sea-bass}$$

If misclassification costs are included, the threshold : $\frac{\lambda_{12}}{\lambda_{21}} \cdot \frac{P(\omega_2)}{P(\omega_1)}$

Likelihood, Threshold, Prior Ratio:

For two classes ω_1, ω_2

Bayes decision rule,

decide ω_1 if $P(\omega_1|x) > P(\omega_2|x)$

Using Bayes Theorem:

$$\frac{P(\omega_1|x)}{P(\omega_2|x)} = \frac{P(x|\omega_1)}{P(x|\omega_2)} \cdot \frac{P(\omega_1)}{P(\omega_2)}$$

So, decide ω_1 if

$$\frac{P(x|\omega_1)}{P(x|\omega_2)} > \frac{P(\omega_2)}{P(\omega_2)}$$

- Likelihood ratio $\frac{P(x|\omega_1)}{P(x|\omega_2)}$ is compared to a threshold.
- The threshold equals the inverse prior ratio $\frac{P(\omega_2)}{P(\omega_1}$
- If $P(\omega_1)$ is larger, the threshold becomes smaller → easier to choose ω_1.